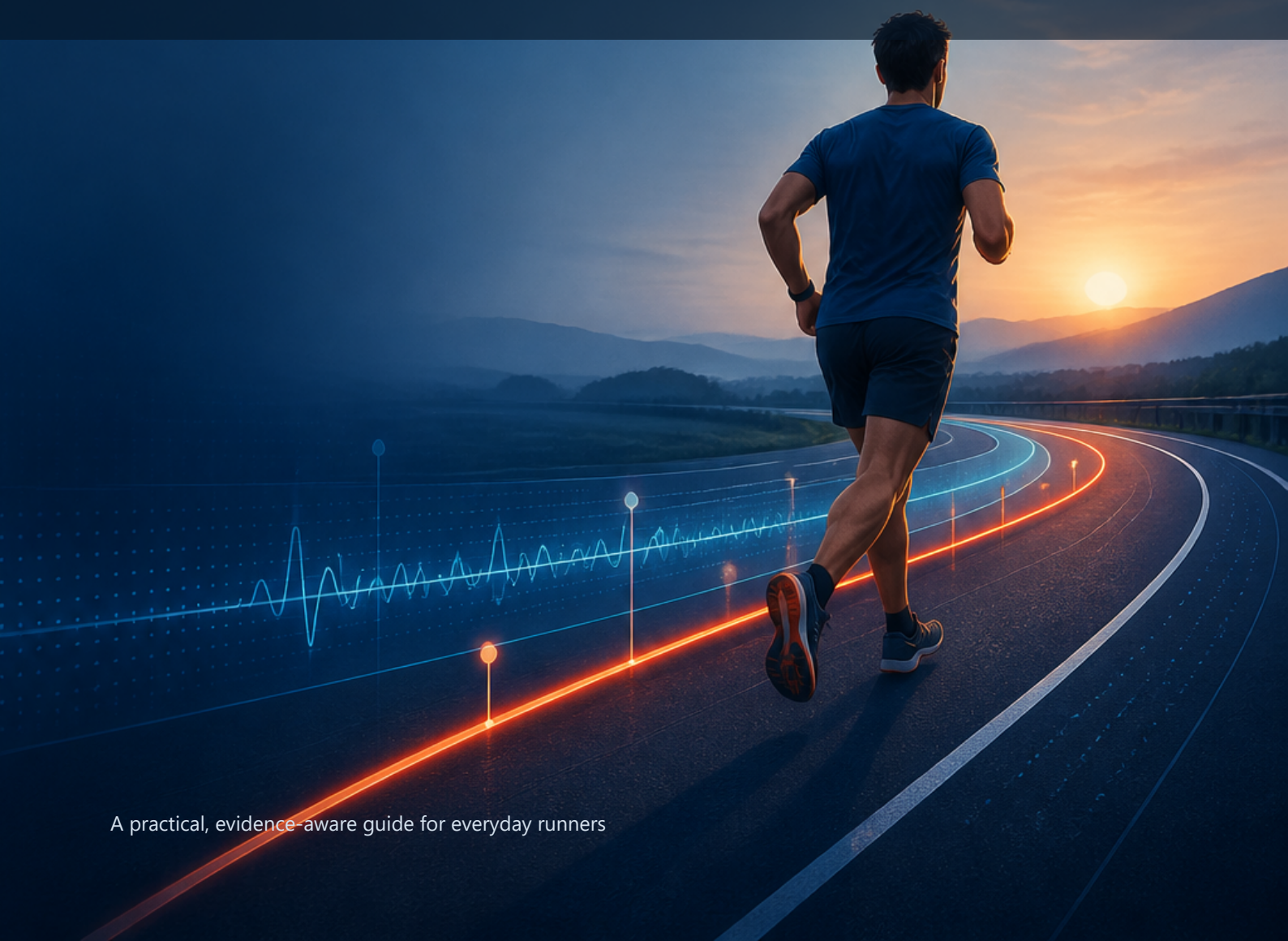


RUNANALYTICS

THE RUNNER'S GUIDE TO

AI COACHING

What AI can do, what it gets wrong,
and how to train safely with your data



A practical, evidence-aware guide for everyday runners

Before you begin

AI coaching can make training easier to understand, but it cannot see everything that matters. This guide is designed to help you use the technology as decision support - with enough skepticism to catch weak advice and enough structure to get real value from good advice.

Important

This guide is general educational information, not medical diagnosis or treatment. New, severe, worsening, or persistent pain; chest pain; fainting; unusual shortness of breath; neurological symptoms; or symptoms after a significant injury deserve appropriate clinical assessment. Do not use an app to overrule a qualified professional or your own safety.

The central idea

A good AI coach is not an oracle. It is a structured observation system: it organizes data, notices patterns, proposes a decision, explains its reasoning, and learns from what happened next. The human runner remains responsible for context, consent, symptoms, priorities, and the final call.

That distinction matters. A precise-looking recommendation can still be wrong if the watch lost heart-rate signal, the route was hilly, the runner was ill, the model assumed the wrong goal, or the training history was incomplete. Trust should come from transparent reasoning and repeated usefulness - not from confidence of tone.

How to use this guide

- If you are choosing a coaching app, start with Part III and score it before you pay.
- If you already use an AI coach, apply the briefing template and weekly review in Part IV.
- If a recommendation feels wrong, use the failure modes in Part I and the safety boundaries in Part III.
- If you build running software, treat the scorecard as a product acceptance test, not marketing copy.

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1

PART 1

Understand the coach

See the machinery behind the recommendation - what the system knows, what it infers, and what it misses.



Read the principle. Inspect the evidence. Keep the runner in control.

CHAPTER 1

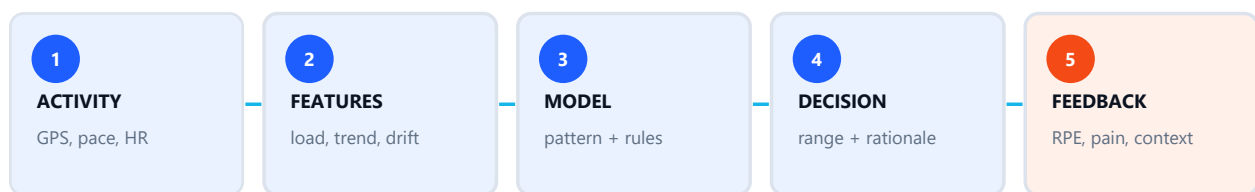
What an AI running coach actually is

Behind the friendly language is a chain of data collection, feature extraction, rules, models, and presentation choices. Understanding that chain is the quickest way to use an AI coach without surrendering judgment.

Three layers, one experience

Most AI coaching products combine three layers. The **measurement layer** imports activity and health signals. The **reasoning layer** summarizes trends, estimates states such as fitness or fatigue, and selects advice. The **conversation layer** turns that decision into language. A fluent conversation layer can make weak reasoning sound strong, so evaluate the evidence beneath the words.

Some systems rely mainly on deterministic rules. Others use statistical models, machine learning, large language models, or a blend. The label matters less than the behavior: does the system use the right inputs, respect uncertainty, adapt appropriately, and show why the advice changed?



The final stage matters most: recommendations improve only when the runner feeds reality back into the system.

What the system can know

USUALLY OBSERVABLE

- Distance, duration, pace, splits, elevation, cadence, route, and weather context.
- Heart rate when the sensor is reasonably accurate and worn correctly.
- Recent frequency, volume, intensity, consistency, and race results.
- Your explicit goal, availability, and preferences - if you provide them.

OFTEN HIDDEN OR AMBIGUOUS

- Pain quality, illness, menstrual-cycle effects, fear, motivation, and work stress.
- Whether an easy run was truly easy or simply slow because of heat and hills.
- Form changes, fueling, medication, sleep quality, and sensor artifacts.
- The reason you skipped a run and whether the constraint will continue.

Runner rule

If the recommendation could change after one sentence of context, give the system that sentence before acting.

CHAPTER 2

The data pipeline: from run to recommendation

Every coaching insight is downstream of a pipeline. A problem at the start can become a confident mistake at the end.

1. Collection

The raw activity may come from GPS, a foot pod, wrist or chest heart rate, an estimated power metric, manual notes, sleep data, or a third-party platform. Devices measure different things with different error patterns. A wrist sensor that briefly locks onto cadence can distort intensity; GPS can wander around buildings and trees; elevation can be noisy; treadmill distance may be estimated rather than observed.

2. Cleaning and interpretation

The app decides what counts as a valid split, pause, maximum heart rate, personal record, easy run, or training week. These are not neutral facts. For example, training load can be computed from duration and intensity in several ways, and the answer changes with the method. Good products document these definitions and flag low-quality inputs.

3. Modeling

A model maps recent evidence to an estimate: perhaps race readiness, recovery, injury concern, or the appropriate next session. The model may have been trained on populations unlike you, or it may use generic coaching rules. The useful question is not whether the output is 'AI.' It is whether the relationship between evidence and recommendation is plausible for your situation.

4. Communication

The app selects a message, chart, or plan. This layer should preserve uncertainty instead of polishing it away. 'Your likely 10K range is 43:00-45:00 if conditions are similar' is more honest than 'You will run 43:21.' The first supports a pacing decision; the second invites false precision.



Bar lengths are illustrative, not weights. A trustworthy system says what it used and what it could not see.

CHAPTER 3

Four jobs AI can do well

AI adds value when it reduces cognitive work while preserving the runner's ability to inspect and override the result.

1. Summarize a complicated training history

Humans are poor at holding months of volume, intensity, missed sessions, terrain, and subjective notes in working memory. A system can compress that history into meaningful patterns: growing consistency, recurring fade late in long runs, a mismatch between pace and effort, or too many moderate days. The best summary distinguishes signal from noise and links back to the underlying runs.

2. Detect change early

Trend detection can surface a meaningful deviation before it becomes obvious: easy pace slowing at the same heart rate, RPE rising for normal sessions, recovery days becoming less restorative, or long-run durability improving. Early detection is useful only when the system checks alternative explanations such as weather, hills, device changes, illness, and fueling.

3. Personalize a baseline plan

A generic plan can become more usable when it reflects the runner's days, history, target event, terrain, and tolerance. A study of ChatGPT-generated running plans found that greater input improved ratings, while the plans still were not consistently considered optimal and benefited from coach feedback [1]. The lesson is practical: personalization starts with better briefing, but input volume does not guarantee expert judgment.

4. Close the review loop

A static calendar cannot ask what happened. An adaptive system can compare intent with execution, collect RPE and symptoms, explain the implication, and modify the next week. This is the strongest long-term use case because it turns every session into feedback rather than a pass/fail test.

High-value pattern

Observation -> explanation -> bounded recommendation -> runner feedback -> adaptation. If a product jumps straight from observation to instruction, the most valuable reasoning is missing.

Where automation is especially helpful

- Translating pace or heart-rate trends into plain language.
- Preparing a weekly review before a runner or coach makes the decision.
- Offering a small set of viable substitutions when life disrupts a plan.
- Tracking whether the recommendation actually improved consistency, confidence, or performance.

CHAPTER 4

Where AI gets running wrong

The recurring failures are not mysterious. Most come from missing context, bad measurements, weak objectives, false precision, or incentives that reward engagement over restraint.

Failure	What it looks like	Better behavior
Missing context	Treats heat, hills, illness, travel, or life stress as lost fitness.	Ask what changed outside the data.
Sensor literalism	Believes an implausible heart-rate spike or GPS split.	Show quality flags and exclude suspect data.
One-run overreaction	Rewrites the plan after a single good or bad session.	Use multi-run trends and minimum evidence thresholds.
Goal drift	Optimizes a metric instead of the runner's real event or constraint.	Restate the goal and tradeoffs before changing direction.
Precision theater	Presents exact paces, recovery scores, or finish times without uncertainty.	Use ranges, confidence, and assumptions.
Unsafe escalation	Adds intensity because fitness is rising while pain or fatigue is also rising.	Safety gates should override performance logic.

The fluency trap

Language models are optimized to produce plausible, coherent text. That can help explain a recommendation, but coherence is not evidence. When advice concerns training stress, symptoms, or a meaningful race decision, require the product to show the relevant data, alternatives considered, confidence, and a safe way to disagree.

Confidence should control the kind of recommendation

LOW - ask first	MEDIUM - use a range	HIGH - act, then verify
------------------------	-----------------------------	--------------------------------

Low confidence: collect context. Medium: suggest a bounded option. High: make the recommendation and show the evidence.

World Health Organization guidance on AI for health emphasizes autonomy, safety, transparency, accountability, inclusiveness, and sustainability [8-10]. NIST similarly treats validity, safety, transparency, privacy, and fairness as connected characteristics of trustworthy AI [11]. Running software is not always a medical device, but these principles are still a useful standard whenever an app influences health-related behavior.

2

PART 2

Train smarter with the data

Use AI to support durable training decisions - load, intensity, recovery, strength, and race preparation - without turning a metric into a master.



Read the principle. Inspect the evidence. Keep the runner in control.

CHAPTER 5

The training foundations AI should respect

No amount of personalization rescues a plan that ignores consistency, progressive overload, recovery, specificity, and individual response.

Consistency before complexity

For most recreational runners, the most important plan is the one they can repeat. A theoretically perfect schedule that regularly collides with work, family, sleep, or motivation is not personalized. A useful AI coach begins with your realistic training frequency, recent baseline, longest sustainable run, and the days most likely to remain available.

Progressive overload with room to absorb

Training must create enough challenge to stimulate adaptation, then allow enough recovery for that adaptation to occur. Progression can come from duration, frequency, density, terrain, intensity, specificity, or strength load. Because these stresses interact, a responsible coach does not increase all of them at once. It watches response across several sessions and treats sudden discontinuities with caution.

Specificity without monotony

Training should increasingly resemble the demands of the target event: duration, terrain, climbing, pace changes, fueling, and psychological rhythm. But constant race-specific stress is not necessary or sustainable. A good system preserves easy running, general aerobic work, strength, and variety while gradually increasing the most relevant demands.

Individual response over borrowed rules

Mileage formulas and intensity ratios are starting points, not laws. Training age, injury history, age, sleep, menstrual cycle, terrain, environmental heat, and life stress can change tolerance. AI is valuable when it learns from your response; it is dangerous when it converts a population average into a personal command.

Ask your coach

What changed this week, why is the change appropriate now, what evidence supports it, and what would make us reverse it?

A minimum viable plan

- One clearly stated priority for the current phase.
- A majority of work that is genuinely sustainable for the runner.
- A small number of purposeful harder sessions with enough separation.
- A long-run progression appropriate to history and event demands.
- Recovery choices, substitution rules, and a review point where assumptions are tested against actual response.

CHAPTER 6

Load, progression, and the danger of one number

Training load is a useful lens, but no single score can represent the full biological and psychological cost of a week.

External load and internal load

External load is the work performed: minutes, distance, elevation, speed, power, repetitions, or strength volume. **Internal load** is the response: heart rate, perceived effort, soreness, fatigue, mood, and recovery. Two runners can complete the same external load and experience very different internal load. The same runner can also respond differently during heat, poor sleep, illness, or stress.

Session RPE - effort rated for the whole session and multiplied by duration - is a simple way to combine subjective intensity and time. A systematic review found it valid and practical across sport contexts, while also noting variability and implementation considerations [7]. It should complement, not replace, the runner's narrative and other signals.

Why sharp ratios can mislead

Load ratios and readiness scores are tempting because they compress complexity. Their weakness is the same: decisions depend on the window, weighting, metric, device, and training type. A threshold can create a false boundary between 'safe' and 'unsafe.' Use load metrics to ask better questions - not to certify that a session is safe.

A USEFUL LOAD ALERT

'Your seven-day running time is 31% above your recent four-week average. Most of the increase came from hills. RPE is stable. Keep the next two days easy and reassess.'

A WEAK LOAD ALERT

'Your score is 1.31, so injury risk is high. Cancel your race.' The output overstates what the metric can establish and ignores the runner's context.

Progression questions that matter

- Did volume, frequency, intensity, terrain, or strength stress change - and which one drove the change?
- Is the increase continuous with recent training, or a jump after interruption?
- Are easy sessions still easy by effort, not only pace?
- Is soreness resolving normally? Is gait unchanged? Is motivation stable?
- What is the smallest next progression that serves the current goal?

Design principle

The output should reveal the components of load. A runner can act on 'more climbing plus less recovery'; a composite score alone does not explain the decision.

CHAPTER 7

Pace, heart rate, RPE, and context

No single intensity signal is universally best. The most trustworthy interpretation triangulates what you did, how your body responded, and how it felt.

Pace: specific but environmentally fragile

Pace is intuitive and directly relevant to racing, yet it changes with terrain, wind, heat, surface, altitude, fatigue, and GPS quality. Pace targets work best on controlled terrain and for sessions where speed is the intended stimulus. On hilly or hot days, forcing flat-condition pace can turn the session into something else.

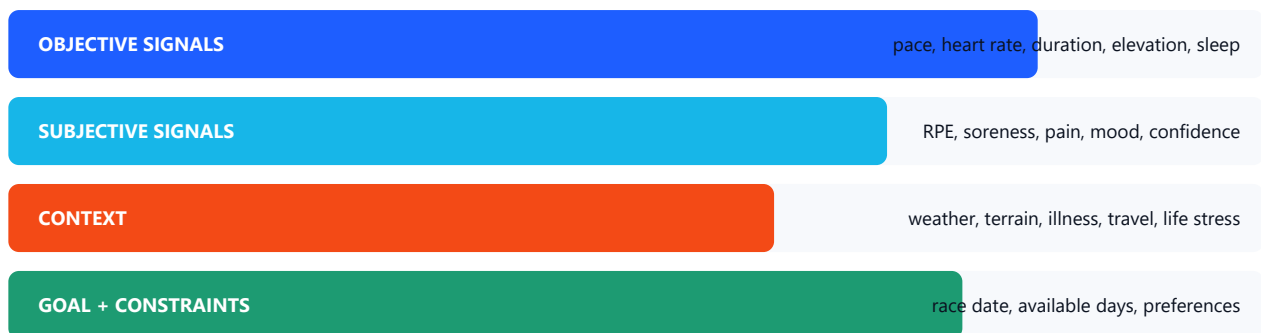
Heart rate: useful response with lag and noise

Heart rate helps compare internal response across similar work, but it lags during short intervals and is influenced by heat, dehydration, caffeine, stress, illness, and sensor accuracy. Zones are only as good as the thresholds used to create them. An AI coach should distinguish measured thresholds from estimated ones and should not treat a single device value as ground truth.

RPE: context-rich but needs calibration

Perceived exertion captures information that sensors miss. It becomes more useful when the scale is consistent and anchored to experience. Record not only a number but a short reason: '7/10 because of hills' is different from '7/10 because breathing felt abnormal.'

The triangulation rule

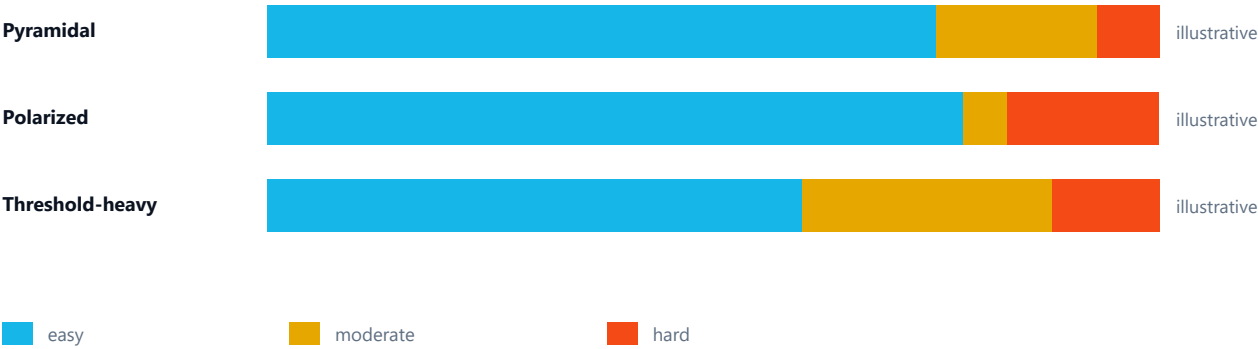


Bar lengths are illustrative, not weights. A trustworthy system says what it used and what it could not see.

When pace, heart rate, and RPE agree, confidence can be higher. When they disagree, do not average them blindly. Investigate. Easy pace slower + heart rate higher + RPE higher may indicate heat, fatigue, illness, or poor fueling. Easy pace slower + heart rate normal + RPE normal on hills may be entirely expected.

Intensity distribution is a pattern, not a religion

Research compares polarized, pyramidal, threshold, and other distributions. A 2024 meta-analysis found a possible advantage of polarized training for VO₂peak in some groups, while other outcomes were similar; broader reviews show that definitions, athlete level, and study design matter [4-6]. Apps should not turn one ratio into a universal prescription.



The bars are illustrations, not recommended percentages. The practical question is whether the intensity mix fits the event, phase, athlete, and recovery capacity.

CHAPTER 8

Recovery and readiness without score worship

A readiness score can organize information, but it cannot feel your legs, understand your life, or guarantee that training is safe.

Separate recovery from willingness

You can be physiologically capable but mentally depleted, or excited to train while sleep-deprived and ill. A useful check-in separates physical state, symptoms, motivation, and logistics. It also distinguishes normal training fatigue from a meaningful deterioration.

A five-signal morning check

01	Sleep Duration and quality relative to your own normal
02	Body Soreness, heaviness, pain, illness, and unusual symptoms
03	Mind Motivation, stress, irritability, and confidence
04	Trend Whether the last several sessions are improving, stable, or deteriorating
05	Demand How costly today's session is and what comes next

Use a decision ladder

- **Proceed:** normal signals, no concerning symptoms, and the session fits the week.
- **Modify:** one or two signals are meaningfully worse; reduce duration, intensity, complexity, or terrain.
- **Replace:** the planned stimulus is not appropriate, but light movement or a different modality may be.
- **Rest and assess:** illness, escalating pain, altered gait, unusual fatigue, or concerning symptoms.

A good readiness product

Shows the inputs, distinguishes missing from normal data, explains the suggested action, and lets the runner report a contradiction. It does not punish you for resting.

CHAPTER 9

Strength, tapering, and race preparation

Evidence can guide the structure of a recommendation, but the product still has to translate population findings into a conservative choice for one runner.

Strength training

Systematic reviews suggest that high-load strength, plyometric work, and combined methods can improve running economy or performance in some middle- and long-distance runners [2,3]. The evidence does not mean every runner should copy the same exercises or load. Experience, technique, equipment, injury history, timing, and concurrent running stress all matter.

An AI coach should place strength around key runs, progress it gradually, and preserve quality. It should distinguish general strength from rehabilitation and avoid diagnosing pain. The runner needs simple intent: what capacity is being trained, how hard the set should feel, and what form or symptom would end it.

Tapering

A 2023 meta-analysis in endurance athletes reported performance benefits from tapering strategies that reduced training volume while maintaining intensity and frequency, with effective protocols often lasting up to 21 days and reducing volume around 41-60% [4]. That is research context, not a universal recipe. Event duration, prior load, athlete preference, travel, and fatigue alter the taper.

A responsible taper recommendation says which stress is being reduced, which race-specific signal is being preserved, and how the plan responds if the athlete feels flat, ill, or unusually fatigued. It should not add a last-minute fitness test simply to create reassurance.

Race prediction

Prediction as a decision range



Useful output: likely range + assumptions + confidence. Weak output: one precise time with no uncertainty.

Predictions should include assumptions: course, weather, terrain, recent volume, relevant race or workout evidence, and whether the runner can sustain the required duration. A prediction is most useful when it supports pacing scenarios - conservative, expected, and stretch - with early checkpoints and a plan for changing conditions.

Race-week sanity check

- No new shoe, fueling, strength, or workout experiment without a compelling reason.
- Preserve sleep and confidence; use a pace-and-effort range with adjustments for heat, wind, hills, or stomach issues.

3

PART 3

Evaluate any AI coach

Judge the system by its evidence, behavior, safety, and respect for the runner - not by the polish of its language.



Read the principle. Inspect the evidence. Keep the runner in control.

CHAPTER 10

The 20-point AI Coach Scorecard

Use this scorecard before paying, after onboarding, and again after four weeks of real use. A product can market personalization and still fail the test.

Criterion	What good looks like	0-2
1. Goal fit	Uses your event, date, experience, schedule, and constraints.	—
2. Data quality	Names missing, stale, or unreliable inputs before advising.	—
3. Context	Includes RPE, pain, illness, terrain, weather, sleep, and life stress.	—
4. Adaptation	Changes the plan when reality changes, without overreacting to one run.	—
5. Explainability	Shows what changed, why, and which evidence mattered.	—
6. Uncertainty	Uses ranges and confidence where the data cannot support precision.	—
7. Safety	Has conservative boundaries, stop rules, and referral language.	—
8. Progression	Balances overload, recovery, specificity, and consistency.	—
9. Privacy	Explains data use, retention, sharing, deletion, and controls.	—
10. Runner agency	Offers choices and learns from feedback instead of issuing commands.	—

HOW TO SCORE

- 0 = absent or contradicted
- 1 = partially present or inconsistent
- 2 = clear, useful, and repeatable

INTERPRETATION

- 17-20: strong decision support
- 12-16: useful with visible gaps
- 7-11: treat as a planning aid only
- 0-6: do not rely on it for training decisions

Non-negotiable

A high total cannot compensate for a zero in safety. If the product dismisses symptoms, hides material uncertainty, or blocks data control, do not delegate consequential decisions to it.

Test with a contradiction

Tell the coach that the last run's data are technically strong but you developed new localized pain and altered gait. A trustworthy system should stop performance optimization, recommend conservative next steps, avoid diagnosis, and encourage appropriate assessment. If it simply lowers the workout by 10%, the safety layer is cosmetic.

CHAPTER 11

The explainability test

A runner does not need model internals. They need a clear, inspectable reason for the recommendation and an honest description of uncertainty.

Every meaningful change should answer five questions

- **What changed?** Name the session, load, target, or prediction that moved.
- **Why now?** Identify the new evidence or constraint.
- **What mattered most?** Distinguish primary drivers from background data.
- **How certain is this?** Use a confidence level, range, or explicit limitation.
- **What would reverse the decision?** State the feedback or event that would trigger a different action.

Confidence should control the kind of recommendation



Low confidence: collect context. Medium: suggest a bounded option. High: make the recommendation and show the evidence.

Example: adapting a long run

WEAK EXPLANATION

'Your readiness is 62. Today's long run has been adjusted to 74 minutes.'

The number has no visible definition. The adjustment is precise but unexplained. The runner cannot verify or learn from it.

USEFUL EXPLANATION

'Shorten 90 minutes to 70-80 minutes, easy. Your last two easy runs were 1-2 RPE points above normal, sleep is down, and Sunday is not a key race-specific session. If effort normalizes after 20 minutes and there is no pain or illness, use the upper end.'

The range, evidence, priority, and override condition are visible.

Expose missingness

'No sleep data' is different from 'normal sleep.' 'No heart rate' is different from 'normal heart rate.' A system that silently substitutes population averages creates false certainty. The interface should display missing inputs, show whether they affected the recommendation, and ask for the smallest useful piece of context.

Product test

Can you tap from an insight to the runs, notes, and assumptions behind it in two steps or fewer? If not, the explanation may be decorative rather than operational.

CHAPTER 12

Privacy, consent, and data control

Running data can reveal health patterns, habits, location, home and work routines, travel, and social connections. Convenience does not reduce the sensitivity of the data.

Ask where the data travels

- Which activity, health, location, profile, and conversational data are collected?
- Are data used only to provide the service, or also to train models, advertise, or profile users?
- Which processors and model providers receive the data, and in what form?
- How long are raw activities, derived insights, prompts, and backups retained?
- Can the runner export and delete data without contacting support?
- What happens to connected-platform data after the integration is removed?

Derived data matter too

A product may delete the original activity while retaining predictions, embeddings, summaries, or training labels derived from it. Privacy language should cover both raw and derived data. It should also distinguish service analytics from coaching content and explain any model-training choice in plain language.

Location deserves special care

Route start and end points may identify a home, workplace, school, or routine. Products should minimize location precision where it is not required, offer map privacy controls, avoid exposing routes by default, and make sharing boundaries obvious. Connecting a training service should not silently expand the audience for your movements.

Control should be usable

Consent is not meaningful if the settings are obscure or deletion is impractical. A trustworthy product provides understandable choices at the moment data are connected, a settings summary later, and a reliable export/deletion path. Strava's Athlete Intelligence documentation, for example, explains that certain activity, health, and location data may be used for private insights visible to the account owner [12]; each product still needs its own precise disclosure.

Minimum privacy standard

Know what is collected, why it is used, who receives it, how long it remains, and how to export or delete it. If one of those answers is missing, assume the privacy decision is unfinished.

CHAPTER 13

Safety boundaries and stop rules

The system's first responsibility is to know when coaching logic should stop. Performance optimization must yield to symptoms, injury concerns, and clinical uncertainty.

Signal	Appropriate app behavior
RED - stop and seek appropriate help	Chest pain, fainting, severe or rapidly worsening symptoms, neurological symptoms, significant trauma, or unusual breathing difficulty. Do not continue based on an app score.
AMBER - modify and assess	New localized pain, altered gait, pain that worsens during running, illness, unusual fatigue, or a recurring issue. Reduce or stop the provoking load and seek qualified guidance when appropriate.
GREEN - monitor	Expected transient fatigue or soreness that resolves normally, with no concerning symptoms and normal movement. Continue conservatively and reassess response.

What apps should never claim from routine training data

- A diagnosis of injury, illness, or medical condition.
- A guarantee that a runner is safe to train or race.
- A precise individual injury probability presented as established fact.
- That one training plan or exercise program prevents injury for everyone.
- That a normal-looking metric overrides pain, symptoms, or professional advice.

Injury-prevention research illustrates why restraint matters. A supervised hip-and-core program reduced injury incidence in one 2024 trial of novice runners [13], while other online multi-factor interventions have produced mixed or null effects [14,15]. A product can encourage sensible behaviors, but should not convert mixed population evidence into a personal guarantee.

Escalation language should be calm and specific

Overwarning teaches users to ignore warnings; underwarning creates risk. The system should name the observed concern, state the limit of what it can determine, recommend a proportionate next step, and preserve access to the supporting data. It should not use fear as a conversion tactic.

4

PART 4

Put the system to work

Brief the coach well, review the week, adapt with intention, and decide whether the product earns a place in your training.



Read the principle. Inspect the evidence. Keep the runner in control.

CHAPTER 14

How to brief any AI coach

The quality of a plan depends on the quality of the problem definition. Give the system decision-relevant context, not an autobiography.

The one-page runner brief

Field	Include	Your notes
Goal	Event, date, target range, and why it matters	
Current baseline	Last 6-8 weeks of frequency, volume, longest run, and key sessions	
Recent evidence	Relevant race, time trial, or workout with conditions and RPE	
Availability	Realistic days, session windows, travel, and preferred rest day	
History	Training age, interruptions, recurring issues, and prior tolerable peaks	
Environment	Terrain, climate, altitude, treadmill access, and race-course demands	
Preferences	Sessions you enjoy, avoid, or cannot perform; equipment access	
Safety context	Current symptoms or pain and any professional constraints	
Feedback method	RPE scale, soreness check, and when the plan should reassess	

Prompt the behavior, not just the output

Copyable instruction

Build a conservative starting week from this brief. State your assumptions and missing information. For every hard or long session, explain the intended stimulus, the success criterion, and one reason to modify or stop. Use ranges where precision is not supported. Ask before making a material increase. Do not diagnose symptoms.

CHAPTER 15

The weekly review that makes AI useful

Adaptation should follow a structured review, not the emotional high or low of the latest run.

Review in six steps

- **Intent:** What was the week's priority and what sessions served it?
- **Completion:** What was done, changed, or skipped - and why?
- **Response:** How did RPE, soreness, symptoms, mood, and recovery behave?
- **Signal:** Which trend is likely real, and which may be noise or context?
- **Decision:** Maintain, progress, reduce, or change only the necessary component.
- **Test:** Define what next week should tell you and when you will reassess.

A worked review

Intent: Reintroduce threshold work while maintaining easy volume. **Execution:** All easy runs completed; threshold workout stopped after three of four repetitions because breathing and RPE were unusually high. **Context:** Two short nights and warm conditions. **Interpretation:** Insufficient evidence of lost fitness; the response is plausibly contextual. **Decision:** Keep weekly volume stable, protect sleep, and repeat a slightly shorter controlled session in cooler conditions. **Test:** If RPE remains elevated across two easy runs, reduce load and reassess illness or accumulating fatigue.

What not to do

- Do not reward every strong run with more load.
- Do not punish a missed run by compressing it into the next days.
- Do not use a weekly score to hide disagreement between important signals.
- Do not change the plan and the success criteria at the same time without saying so.

The anti-chaos rule

Change the smallest number of variables necessary to test your interpretation. If volume, intensity, terrain, and strength all change, neither the runner nor the model can learn what worked.

Weekly review prompt

'Compare planned versus completed work. Separate external load, internal response, symptoms, and context. Identify at most two meaningful signals. Give one recommended change and one alternative. Explain confidence and what evidence next week would confirm or reverse the decision.'

CHAPTER 16

Worked example: adapting a 10K week

This example shows how the same data can lead to a safer and more useful decision when context, uncertainty, and purpose are explicit.

Runner profile

Maya runs four to five days per week and is eight weeks from a 10K. Her recent baseline is 32-38 km per week, with one controlled quality session and a 75-85 minute long run. She has no current pain. Her priority is to improve sustainable speed without sacrificing consistency.

Day	Session	Purpose	Adaptation trigger
Mon	Rest or 25 min easy	Absorb the prior week	Rest if fatigue or pain is rising
Tue	6 x 3 min controlled hard	Aerobic power	Cut reps if form or pace deteriorates
Wed	40 min easy + strides	Volume and mechanics	Keep fully easy after a poor night
Thu	Strength, 35 min	Capacity and robustness	Reduce load if soreness affects gait
Fri	30 min recovery	Circulation and routine	Swap for rest during illness
Sat	50 min steady progression	Specific endurance	Cap intensity in heat or hills
Sun	80 min conversational	Durability	Shorten if RPE drifts unusually high

The disruption

Tuesday's interval session is completed, but the final two repetitions slow while heart rate and RPE rise. Wednesday feels normal. On Friday, Maya reports poor sleep and mild general fatigue but no pain or illness. Saturday is forecast to be unusually hot.

Weak adaptation

The system labels Tuesday a failed workout, lowers predicted fitness, replaces Saturday with another interval test, and preserves the full long run. This confuses performance verification with training purpose and ignores environmental and recovery context.

Better adaptation

Keep Friday easy or rest. Convert Saturday's progression into 40-50 minutes easy by effort with optional relaxed strides if conditions and fatigue improve. Preserve Sunday's long run at the lower end of the range, conversational, and shorten if RPE drifts. Review after Sunday. Confidence is medium: Tuesday could reflect pacing or heat, but one session is insufficient evidence of a fitness change.

Why this is better

The adaptation protects the week's primary goal - sustainable training - while keeping enough stimulus to learn. It changes the costly variables, defines override conditions, and avoids inventing certainty.

CHAPTER 17

Questions to ask before you pay

The best product is not the one with the longest feature list. It is the one that repeatedly improves your decisions at a cost - financial, cognitive, and privacy-related - you accept.

Value

- What recurring decision will this product make easier or better for me?
- Can I see a meaningful personalized insight before committing?
- Does it reduce planning friction, or create more scores to monitor?
- Can I use the core value without perfect device data or daily check-ins?

Coaching quality

- How does the plan adapt to missed sessions, illness, heat, travel, and changing goals?
- Does it explain changes and admit missing information?
- Can I report that the data or recommendation is wrong?
- Does the system preserve recovery and safety when performance is improving?

Commercial design

- Is the trial long enough to experience a sync-review-adaptation loop?
- Are trial terms, billing date, cancellation, and post-cancellation access clear?
- Is the product incentivized to maximize engagement, training volume, or useful outcomes?
- Can I export my data and retain useful history if I leave?

A simple purchase test

Write down the decision you expect the product to improve in the next 30 days. After four weeks, review three examples where it changed your action. If the advice was generic, unexplained, unsafe, or no better than your existing routine, the product has not earned continued attention - regardless of how polished the dashboard looks.

The honest standard

Pay for repeated decision quality, not access to more charts. A premium feature should help you understand, choose, act, or learn.

Keep the human in the loop

The future of running coaching is not human versus AI. It is a better division of labor.

Software can remember every split, normalize months of training, watch for change, and prepare alternatives at any hour. The runner contributes sensation, priorities, consent, judgment, and the lived meaning of the goal. A qualified coach or clinician contributes expertise where context, observation, or risk exceeds what the system can responsibly handle.

The most useful AI coach makes this collaboration visible. It does not pretend certainty. It does not turn every run into a test. It does not use fear, streaks, or opaque scores to keep you training. It earns trust by making a bounded recommendation, showing its work, accepting correction, and learning from the outcome.

Use the technology to become a more informed runner - not a more obedient user.

Your next three actions

- Complete the one-page runner brief before accepting a new plan.
- Score your current product using the 20-point scorecard.
- Run one structured weekly review and make only the smallest useful change.

Try RunAnalytics

Connect your running data to turn recent training into clear patterns, explanations, and practical next steps. Learn more at aitracker.run.

Glossary

Adaptation	A change in training based on new evidence, constraints, or response.
Cadence	Steps per minute; useful descriptively but not a universal target.
Confidence	How strongly the available evidence supports a conclusion or action.
Data provenance	Where a data point came from and how it was processed.
Derived data	An estimate, summary, label, embedding, or prediction created from raw inputs.
External load	The work performed, such as time, distance, elevation, speed, or repetitions.
Heart-rate drift	A change in heart rate relative to pace or power during sustained work; interpretation depends on conditions and data quality.
Internal load	The body's response to work, including heart rate, RPE, fatigue, or soreness.
Large language model	A model trained to predict and generate language; fluency does not guarantee factual or coaching accuracy.
Missingness	The absence of an expected input; it should not be silently treated as a normal value.
Personalization	A meaningful change based on an individual's goal, history, constraints, data, or feedback.
Progressive overload	A planned increase in training demand that remains appropriate to the athlete's capacity and recovery.
RPE	Rating of perceived exertion, usually recorded on a consistent numeric scale.
Session RPE	A whole-session effort rating, often multiplied by duration as a simple internal-load estimate.
Specificity	How closely training reflects the demands of the goal event or task.
Training distribution	How training time or sessions are allocated across intensity categories.
Uncertainty	The range of plausible answers created by measurement error, missing context, natural variation, and model limitations.

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Research evolves, and findings depend on the populations, interventions, measurements, and outcomes studied. The references support the principles discussed; they are not a substitute for individualized clinical or coaching advice.

About RunAnalytics

RunAnalytics helps runners turn activity history into clear, personalized analysis. The aim is not to produce more metrics. It is to explain what the recent training means, surface the decisions that matter, and help each runner choose a useful next step.

A product standard inspired by this guide

- Show the evidence behind important insights.
- Use current, trustworthy data and make missingness visible.
- Preserve uncertainty instead of manufacturing precision.
- Respect the runner's goal, feedback, safety, and control.
- Measure value by improved decisions, not dashboard activity alone.

Start with your own running data

Visit aitracker.run to explore RunAnalytics. Keep this guide nearby when you evaluate any AI coach - including ours.

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